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Climate MD: Bridging Climate Science and Patient Health

Abstract

When Hurricane Maria struck Puerto Rico in 2017, it put the supply chain for intravenous saline out of commission, causing national shortages of a backbone of basic medical care. Climate MD, a Harvard Chan C-CHANGE program, helps doctors, nurses, and other providers understand how climate change affects their patients and their ability to deliver care, and empowers them to act.

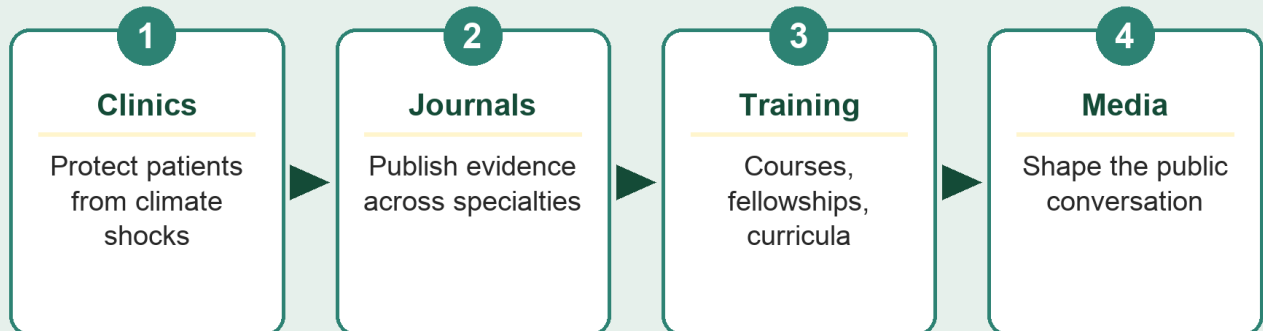
Introduction

Hurricanes, wildfires, extreme heat, and other consequences of burning fossil fuels threaten more than our health directly. They also threaten the ability of medical providers to care for everyone who needs it. Supply chains break, clinics lose power, and patients with chronic conditions face new dangers. Health care providers are trusted voices and see these impacts firsthand, so they have a special role to play. Harvard Chan C-CHANGE created Climate MD with a clear mission: bring climate solutions to the bedside. The program engages, supports, and empowers medical professionals to turn awareness of climate health threats into actions that improve health.

How Do Scientists Study This?

Climate MD works through four strategies. First, it partners with frontline health clinics, working with providers, staff, and patients to protect patients from climate shocks such as heat waves, hurricanes, floods, and wildfires. Second, it publishes and amplifies research in medical journals across specialties, showing connections between climate change and pediatrics, primary care, psychology, emergency medicine, surgery, internal medicine, cancer, lung disease, and infectious diseases. Third, it trains medical leaders through online courses, fellowships, and medical school curricula built on clinical cases. Fourth, it works with the national media as a go-to source on climate and health, advising on how to communicate these issues effectively.

Climate MD's four strategies



Support clinics

Work with frontline clinics to protect patients from heat, storms, floods, and wildfires.

Publish evidence

Share research on climate and health in medical journals across many specialties.

Train and speak out

Prepare medical leaders and help providers use their voices in the media and public.

What Did They Discover?

Climate MD has reached a large and global audience. Its team co-hosted a massive open online course, Health Effects of Climate Change, taken by more than 100,000 students in over 100 countries. It helped run Harvard's Climate Change and Health Fellows program for emergency medicine physicians and the Climate Health Organizing Fellows program for U.S. health professionals. The team outlined the first residency curricula to prepare doctors for climate change and has identified climate and health initiatives at educational institutions. Its work with frontline clinics centers on patient-centered climate resilience and health equity, and its media work helps shape how the public understands the health side of climate change.

Climate MD's reach

100,000+

students took the Health Effects of Climate Change course

100+

countries reached by the online course

9+

medical specialties linked to climate in published work

Climate MD's work spans at least nine medical specialties, showing that climate change touches nearly every area of patient care.

Takeaways

Climate disrupts care

Disasters can break supply chains, as Hurricane Maria did for IV saline.

Every specialty

From pediatrics to cancer care, climate change affects how doctors treat patients.

Providers as voices

Trained clinicians can protect patients and lead public action on climate.

Key Terms

Supply chain

The network of manufacturers, transport, and suppliers that delivers products, such as medical supplies, to where they are needed.

Intravenous (IV) saline

A sterile salt-water solution delivered into a vein, used in basic medical care for fluids and medicines.

Frontline clinic

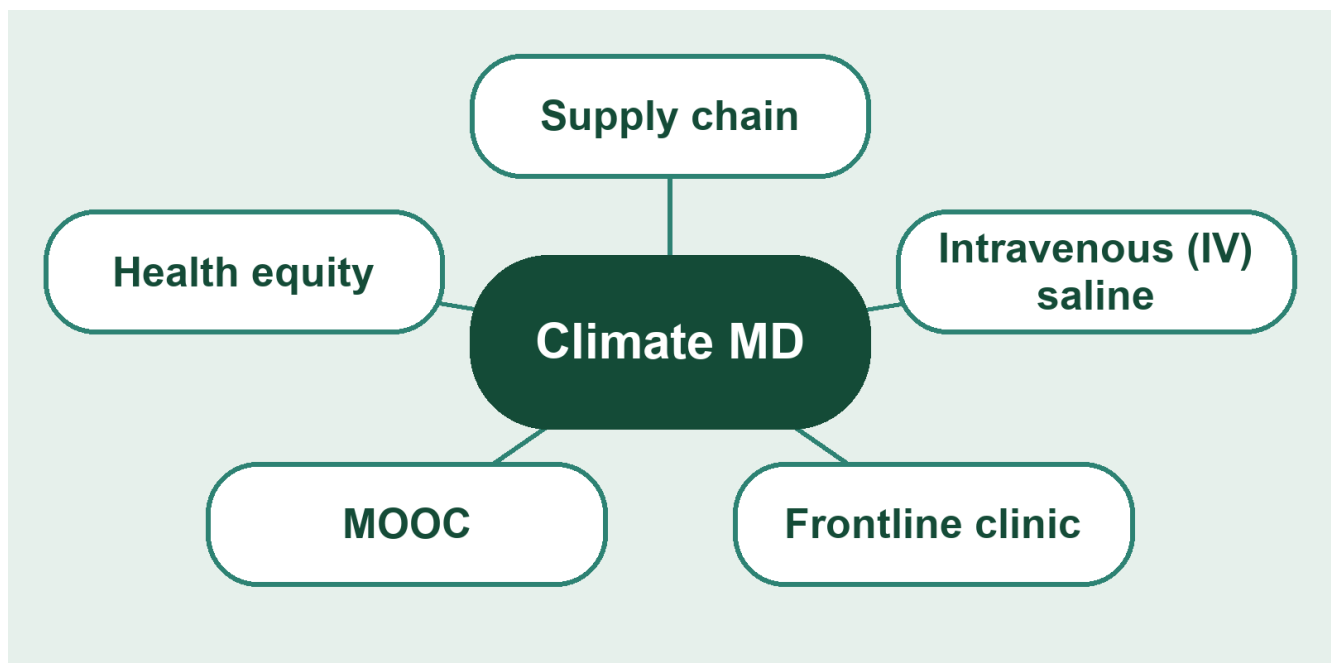
A clinic serving communities most exposed to health and climate risks, such as community health centers.

MOOC

A massive open online course that anyone can take over the internet, often free.

Health equity

The goal that everyone has a fair chance to be as healthy as possible, regardless of income, race, or where they live.



Activity

Bedside Climate Conversation

Students role-play a provider explaining a climate-related health risk to a patient and giving practical advice.

Materials: Patient scenario cards: older adult during a heat wave, child with asthma during wildfire smoke, patient on dialysis before a hurricane; Advice planning sheet

1. In pairs, choose a scenario card. One student is the provider, the other the patient.
2. The provider explains in plain language how the climate hazard could affect the patient's health.
3. The provider gives two practical steps the patient can take to stay safe.
4. Switch roles with a new scenario.
5. Discuss: why might patients listen to a doctor or nurse about climate risks more than to other sources?

References

Harvard T.H. Chan School of Public Health, C-CHANGE. Climate MD.
<https://hsph.harvard.edu/research/climate-health-c-change/outreach/climate-md/>